

Ganesh Iyer

+41 762708010 omkaar0791@gmail.com [in](#) [LinkedIn](#)

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Education

- Sep 2025–2027 **ETH Zürich**, *Master of Science*,
Robotics, Systems and Control
GPA **5.72/6**
- Nov 2021–May 2025 **Indian Institute of Technology Bombay**, *Bachelor of Technology*,
Mechanical Engineering (with Honors) specializing in Systems and Controls,
Minor in Artificial Intelligence and Data Science
GPA **9.87/10**

Skills

- Languages Python, MATLAB, C++
Software NVIDIA Isaac (Lab, Sim, Gym), MuJoCo, ROS2, Git, GitHub, Docker, L^AT_EX
Libraries Pytorch, NumPy, OpenCV, Pandas, Scikit Learn

Internships

- June 2025 – July 2025 **Reinforcement Learning for Quadruped Robot Locomotion**
[xTerra Robotics, India](#)
 - Researched **quadruped locomotion control** via **guided policy search**, where **model-based control** data guides the **RL policy** during training
 - Collected a **3-minute** reference dataset by running a **model-based** controller on the Svan M2 hardware
 - Trained an RL policy to **imitate** these reference trajectories (**joint positions, velocities, torques**, foot placements) via **trajectory-tracking rewards**
 - Used PPO in **NVIDIA Isaac Gym** with domain randomization for Sim2Real, validated in **MuJoCo**
 - Trained NN-based **state estimator** for base linear, angular velocity and base height estimation from **sensor data buffer**, required by the RL Policy
 - Deployed on Svan M2 **hardware**, achieving **trot gait** (0.7 m/s forward, 0.5 m/s lateral, 1 rad/s yaw)
- May 2024 – Jul 2024 **Voronoi Diagram Based Path Planning for Superquadric Obstacles**
Global Connect Fellowship, Nanyang Technological University | Guide: [Prof Domenico Campolo](#) | [GitHub](#)
 - Conducted a comprehensive survey on **superquadrics** and their role in obstacle-based **path planning**.
 - Developed a novel algorithm to generate **Voronoi diagrams** for superquadrics and applied **Dijkstra's** algorithm on the resulting graph for efficient path planning.
 - Used **image processing** to extract object **point cloud** data from an environment image, fitted superquadrics onto obstacles, and validated on a physical maze of variously-shaped objects
 - Benchmarked the Voronoi-based planner against **classical methods** like A* and RRT*, and **heuristic** planners such as genetic algorithms, and a recent state-of-the-art drone planning algorithm in 3D.
 - Concluded that Voronoi planning for superquadrics offers **safer** paths with fewer nodes, **low computational space**, requiring only object **point cloud** from image, and no other parameters.

Publications

- Conference Paper Lin Yang, **Ganesh Iyer**, Baichuan Lou, Sri Harsha Turlapati, Chen Lv, Domenico Campolo, **"Path Planning in Complex Environments with Superquadrics and Voronoi-Based Orientation"**, IEEE Conference on Robotics and Biomimetics 2025

Projects

- Feb 2026 – Present **Humanoid Motion Control for Soccer**
NomadZ – RoboCup Team, ETH Zürich | nomadz.ethz.ch
 - Developing learned motion skills for **Booster K1** humanoid robots in RoboCup soccer
 - Retargeted custom-recorded human motion (walking, dribbling) onto the robot using **GVHMR** and **GMR**
 - Trained velocity-tracking **steering** and **dribbling** policies using **Adversarial Motion Priors (AMP)** for natural style, in **Isaac Lab** (MimicKit), with sim2sim testing in **MuJoCo**
 - Trained a **fall-recovery RL** policy using keyframe-guided learning, to get up from supine, prone poses
 - Building **Sim2Real** deployment and integrating with the vision and behaviour pipelines

- Spring 2026 **Cloth Folding with a Diffusion Policy**
[Robot Learning Course](#), ETH Zürich | [GitHub](#)
- Assembled a **LeRobot SO-101** arm and collected around **200 teleoperated** demonstrations of folding a square towel in three stages: grasp a corner, first fold, and second fold
 - Trained a **multi-task Diffusion Transformer (DiT)** policy (LeRobot) with **CLIP** vision and text encoders, predicting action chunks from a single wrist-camera RGB feed
 - Deployed on the **real robot** with **asynchronous inference** on a local **RTX 4060 (8GB)**
- Dec 2025 – **Monocular Visual Odometry**
 Jan 2026 Mini Project | [VAMR 2025](#) | [GitHub](#)
- Built a monocular visual odometry pipeline (Python/OpenCV) using **Harris** corners + SIFT, **KLT optical flow** tracking, **PnP** + RANSAC pose estimation, triangulation to recover 6-DoF camera poses
 - Added robust bootstrapping, nonlinear **pose refinement** (Levenberg–Marquardt), re-initialization, and landmark rescaling to handle drift, feature loss, and scale ambiguity
 - Tuned and validated across KITTI, Malaga, Parking, and a custom ETH Polybahn ride dataset, achieving **locally consistent** trajectories. Results can be viewed [here](#).
- Aug 2024 – **Koopman/DMD based Control for Non-Linear Dynamical Systems**
 Apr 2025 Bachelor Thesis Project, IIT Bombay | Guide: [Prof. Abhishek Gupta](#)
- Extensively Studied **Koopman theory**, and many data-driven control strategies, including Dynamic Mode Decomposition(**DMD** and **EDMD**), and its control variants(**DMDc**), and **Hankel-based MPC**
 - Applied Koopman Theory to predict dynamics of simple systems (spring-mass, vanderpol oscillators, pendulum), then extended it to a complex **two-link manipulator**
 - Applied Koopman/DMD on simulation data of a planar soft robot to obtain **approximate linear dynamics** in the **lifted space**
- Aug 2024 – **Warehouse Drone - E-Yantra, Nation-wide Robotics competition**
 Sep 2024 Competition | [eYRC portal](#)
- Developed PID control algorithm for roll, pitch, yaw and throttle control of a Swift Pico Drone in Gazebo, with the node files and communication pipeline in ROS2 environment
 - Used **A*** for **motion planning** to **navigate** and control the drone in a **warehouse** simulation
 - Team ranked **40 among 3000+ teams** across the nation participating in eYRC, and among the top 50 teams selected for the **hardware phase**, to implement our algorithms on a Quadcopter

Scholastic Achievements

- 2024 **Department Rank 2** out of **196** students in Mechanical Engineering, IITB
- 2023 Awarded the **Institute Academic Prize two times**, for exceptional academic performance in the second and third year among Mechanical Engineering students
 Recipient of the **Undergraduate Research Award** for exemplary research work in the second year summer
 Achieved perfect **10** Semester Performance Index in **five semesters**: Semester **3,4,5,7** and **8**
 Awarded the **Academic Proficiency(AP)** grade for **exceptional academic performance** in **7 courses**
- 2021 Secured **All India Rank 179** in **JEE Main** among 1.2 million aspirants
 Secured **All India Rank 1234** in **JEE Advanced** among 250,000 aspirants

Positions of responsibility

- May 2023 – **Teaching Assistant | Introduction to Quantum Physics (PH107)**
 July 2023 Prof Alok Shukla, Department of Physics
- Conducted tutorial sessions for **40+ students** across various departments
 - Covered content and clarified doubts of the students pertaining to **Quantum Physics**
 - Graded the Midsemester and Endsemester answer sheets for the batch and conducted crib sessions
- May 2023 – **Student Mentor | Mechanical Department Academic Mentorship Programme**
 May 2024 Selected as a part of a 35 membered team out of 125+ applicants based on interviews and peer reviews
- Mentored **5 sophomores**, helping them strike a balance between academics and extracurriculars
 - Acting as the first point of contact in order to actively bridge the gap between faculty and students
- May 2024 – **DCAMP Mentor | Mechanical Department, IIT Bombay**
 Oct 2024 An initiative by the Mechanical department to ensure proper career advice for pre-final year students
- Mentored **4 pre-final year students**, helping them get a better perspective of career opportunities
 - Guided students through their internship season ensuring regular **interactions** and **advice**